



Barker College

**2009
TRIAL
HIGHER SCHOOL
CERTIFICATE**

Mathematics Extension 1

Staff Involved:

PM THURSDAY 20 AUGUST

- LJP*
- PJR*
- MRB
- GDH
- WMD
- RMH
- BTP

75 copies

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Make sure your Barker Student Number is on ALL pages of your solutions
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- ALL necessary working should be shown in every question

Total marks – 84

- Attempt Questions 1 – 7
- All questions are of equal value
- Marks may be deducted for careless or poorly arranged working

Total marks – 84

Attempt Questions 1–7

ALL questions are of equal value

Answer each question on a SEPARATE sheet of paper

Marks

Question 1 (12 marks) [START A NEW PAGE]

- (a) Find the coordinates of the point P which divides the interval joining A (-2, 3) and B (3, -4) externally in the ratio 3 : 2 2

- (b) Evaluate $\lim_{x \rightarrow 0} \left(\frac{\sin 5x}{2x} \right)$ 2

- (c) Solve $\frac{3}{2x - 4} > -2$ 3

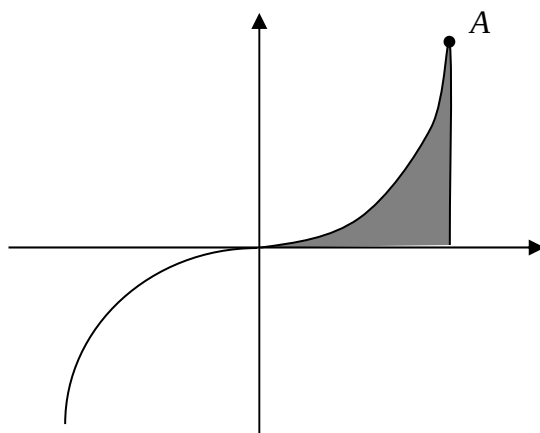
- (d) Evaluate $\int_{-1}^1 \frac{-1}{\sqrt{2-x^2}} dx$ 2

- (e) Evaluate $\int_0^{\frac{\pi}{12}} 2 \sin^2 4x dx$ 3

Question 2 (12 marks) **[START A NEW PAGE]**

(a) Find $\int 6x^3 \sqrt{(3x^4 - 3)^3} dx$ using the substitution $u = 3x^4 - 3$ 2

(b) The graph of $y = 2 \sin^{-1}\left(\frac{x}{3}\right)$ is shown below



(i) Write down the coordinates of Point A 2

(ii) Differentiate $y = 2x \sin^{-1}\left(\frac{x}{3}\right) + 2\sqrt{9 - x^2}$ 2

(iii) Hence, or otherwise, find the shaded area 2

(c) (i) Express $\sqrt{3} \sin t + \cos t$ in the form $R \sin(t + \alpha)$ 2
where α is in radians

(ii) Hence, or otherwise, find the solutions of the equation 2
 $\sqrt{3} \sin t + \cos t = \sqrt{3}$ for $0 \leq t \leq 2\pi$

Question 3 (12 marks) **[START A NEW PAGE]**

- (a) Consider the function $y = (x + 2)^2 + 1$
- (i) Write down the entirely negative domain for which the inverse function exists **1**
- (ii) State the domain of this inverse function **1**

- (b) Find the constant term in the expansion $\left(x^4 + \frac{3}{x^2}\right)^{15}$ **2**

- (c) A hot drink is placed in a closed room, where the temperature is a constant 15°C
The cooling of the drink follows the rule

$$\frac{dT}{dt} = -k(T - 15)$$

where k is a constant, t is the time in minutes and T is the temperature in $^\circ\text{C}$

- (i) Show that $T = 15 + Ae^{-kt}$ satisfies this equation, where A is a constant **1**
- (ii) The hot drink is initially 88°C and cools to 55°C after 11 minutes. **2**
Find the value of the constants A and k , leaving in exact form
- (iii) How long will it take for the drink to cool to 33°C (to the nearest second)? **2**

- (d) Prove, by mathematical induction, that for integers $n \geq 1$ **3**

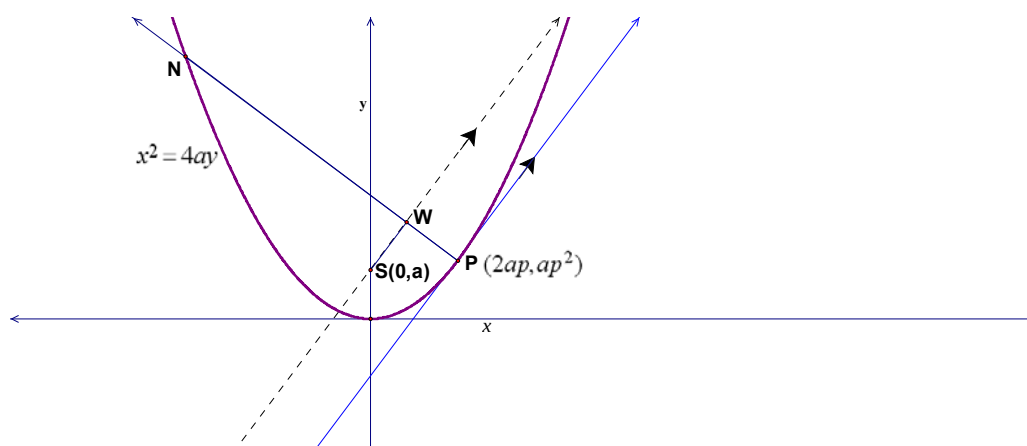
$$1^2 + 3^2 + 5^2 + \dots + (2n - 1)^2 = \frac{1}{3}n(2n - 1)(2n + 1)$$

Question 4 (12 marks) **[START A NEW PAGE]**

- (a) NP is the normal to the parabola $x^2 = 4ay$ at the point $P(2ap, ap^2)$,
 where N also lies on the parabola $x^2 = 4ay$
 W lies on NP such that SW is parallel to the tangent to the parabola at P ,
 where S is the focus of the parabola $x^2 = 4ay$
 You may assume the equation of the normal at P is
 $x + py = ap^3 + 2ap$ (Do NOT prove this)

- (i) Show that the coordinates of the point W are $(ap, ap^2 + a)$ 2

- (ii) Hence, or otherwise, find the locus of W 1



Question 4 continues on page 6

Question 4 (continued)

- (b) A particle is moving so that its displacement, x metres, from the origin is given by

$$x = 2 \cos\left(3t - \frac{\pi}{6}\right)$$

where t is in seconds

- | | | |
|-------|--|---|
| (i) | Show that the motion is simple harmonic | 1 |
| (ii) | Write down the period | 1 |
| (iii) | Find the velocity when the particle is first at $x = \sqrt{3}$ | 2 |

- (c) The acceleration of a particle moving along a straight path is given by

$$\ddot{x} = -2e^{-x}$$

where x is in metres.

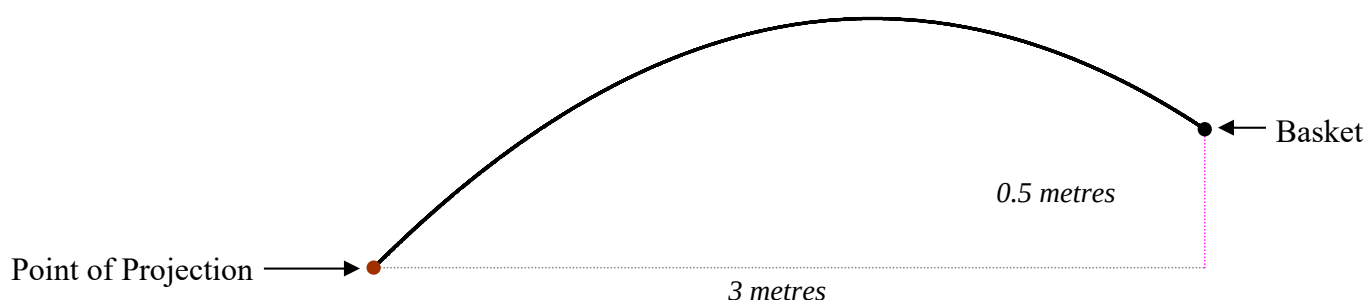
Initially, the particle is at the origin with a velocity of 2 m/s

- | | | |
|------|--|---|
| (i) | Show that $v = 2e^{-x/2}$ | 2 |
| (ii) | Find the equation of displacement, x , in terms of t seconds | 3 |

End of Question 4

Question 5 (12 marks) **[START A NEW PAGE]**

(a)



Greg is about to have a shot at goal in a game of basketball.

From the point where the ball leaves his hand, the distance to the top of the basket is 3 metres horizontally and 0.5m vertically.

Greg shoots at the optimal angle of 45° .

You may assume the equations of motion are

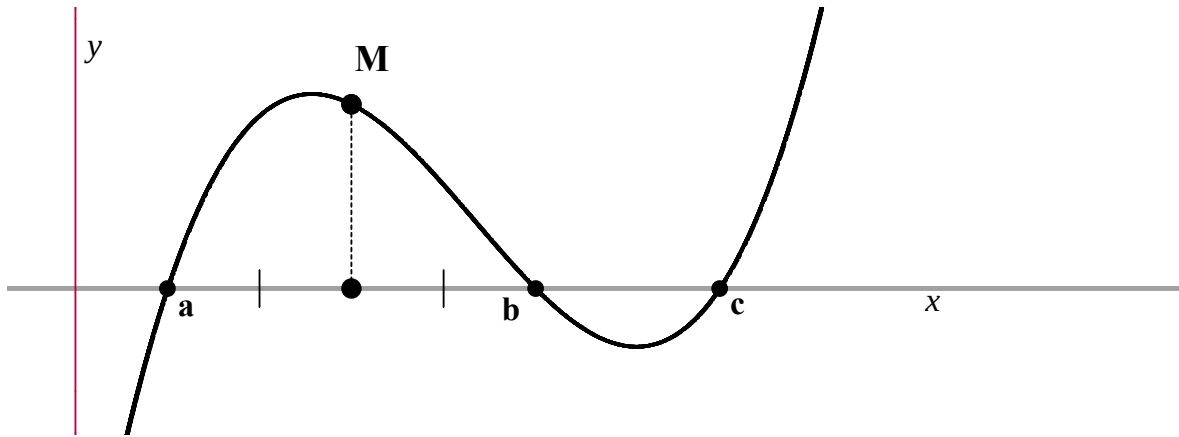
$$x = vt \cos 45^\circ \text{ and } y = vt \sin 45^\circ - 5t^2 \text{ (Do NOT prove this)}$$

- | | | |
|-------|---|----------|
| (i) | Find the velocity of projection, v , required by Greg for the centre of the ball to land in the centre of the basket. | 2 |
| (ii) | Find the maximum vertical height above the basket that the ball reaches during Greg's shot. | 2 |
| (iii) | Find the speed of the ball on entry into the basket. | 2 |

Question 5 continues on page 6

Question 5 (continued)

- (b) The graph below shows the cubic function $f(x) = (x - a)(x - b)(x - c)$ where $x = a$, $x = b$ and $x = c$ are the x -intercepts.



The derivative of this function is given by

$$f'(x) = (x - a)(x - b) + (x - a)(x - c) + (x - b)(x - c) \quad (\text{Do NOT prove this})$$

which is the product rule extended, ie. $y' = uvw' + uwv' + vwu'$

The point M lies on this cubic function $y = f(x)$ such that the x -value of M

is $x = \frac{a+b}{2}$, ie. the mid-value of $x=a$ and $x=b$.

- (i) Show that the coordinates of M are $\left(\frac{a+b}{2}, -\frac{1}{8}(a-b)^2(a+b-2c)\right)$ 2

- (ii) Show that the gradient of the tangent at M is given by 2

$$f'\left(\frac{a+b}{2}\right) = -\frac{(a-b)^2}{4}$$

- (iii) Find the gradient of the straight line passing through M and the point $(c, 0)$. 1

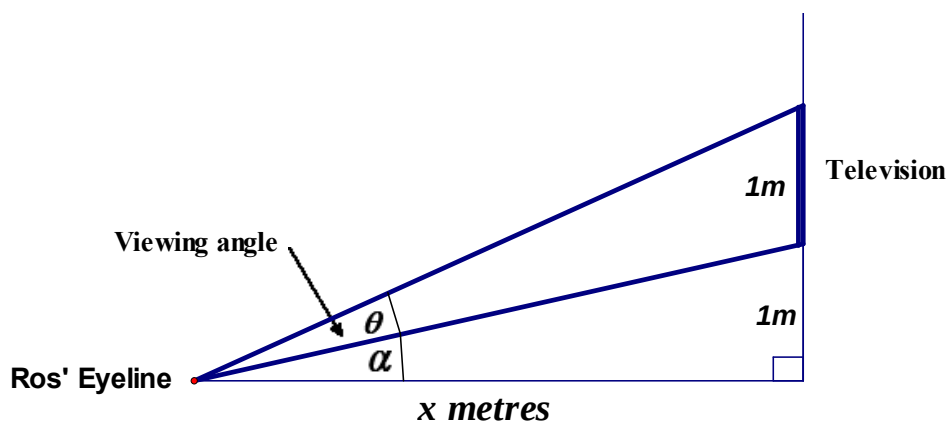
- (iv) What can you conclude about the tangent at M and its x -intercept? 1

Justify your answer.

End of Question 5

Question 6 (12 marks) **[START A NEW PAGE]**

- (a) Ros buys a new 1 metre tall plasma television.
 She mounts it on a vertical wall, placing it so that the base of the television is 1 metre above her (horizontal) eyeline from where she sits in her favourite armchair.
 Let the distance from her eye to the wall be x metres and the angle from her eye to the top and base of the television be θ (the viewing angle).
 Let α be the angle of elevation to the base of the television.

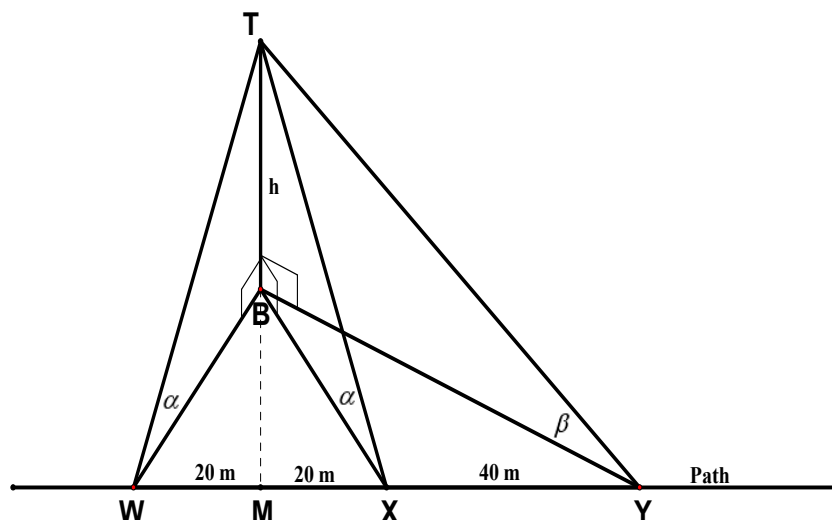


- (i) Show that $\theta = \tan^{-1}\left(\frac{2}{x}\right) - \tan^{-1}\left(\frac{1}{x}\right)$ 2
- (ii) Find the value of x which gives the maximum viewing angle. 3
- (iii) Hence, find the maximum viewing angle of θ , to the nearest degree. 1

Question 6 continues on page 10

Question 6 (continued)

(b)



Three points W , X and Y lie on a straight level path, where $WX = XY = 40 \text{ metres}$.

The base B of a flagpole TB is level with the path. M is the midpoint of WX .

The angles of elevation to the top of the flagpole from the points W , X and Y are α , α and β respectively.

(i) Prove that $\triangle BWX$ is isosceles. 2

(ii) Find the length of BM . 1

(iii) Hence, or otherwise, show that the height of the flagpole is given by 3

$$h = \frac{40\sqrt{2}}{\sqrt{\cot^2 \beta - \cot^2 \alpha}}$$

Question 7 (12 marks) **[START A NEW PAGE]**

- (a) In how many ways can five boys and two girls be arranged in a row if
- (i) there are no restrictions 1
 - (ii) the girls must be together 2
 - (iii) there are exactly two boys separating the girls 2
- (b) Laurence was asked to sketch the graph of the curve $y = \frac{x}{\log_e(x^2)}$
- (i) He was about to change $\log_e(x^2)$ to $2\log_e x$, but then realised this would actually alter the graph itself. Briefly explain why. 1
 - (ii) Accurately describe the domain. 1
 - (iii) Find the derivative of the curve. 1
 - (iv) Find the two turning points of this curve and determine their nature. 2
 - (v) Sketch the graph of this curve clearly labelling all critical features. 2

End of Paper

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Question 1

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$$(a) \left(\frac{kx_2 + lx_1}{k+l}, \frac{ky_2 + ly_1}{k+l} \right)$$

$$\begin{matrix} (-2, 3) & (3, -4) & 3-2 \\ x_1, y_1 & x_2, y_2 & k, l \end{matrix}$$

$$P = \left(\frac{3 \times 3 + (-2) \times (-2)}{3-2}, \frac{3 \times (-4) + (-2) \times 3}{3-2} \right)$$

$$= \left(\frac{9+4}{1}, \frac{-12-6}{1} \right)$$

$$= (13, -18)$$

$$(b) \lim_{x \rightarrow 0} \left(\frac{\sin 5x}{5x} \times \frac{5}{2} \right)$$

$$= 1 \times \frac{5}{2}$$

$$= \frac{5}{2}$$

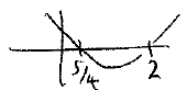
$$(c) (2x-4) \times \frac{3}{(2x-4)^2} > -2(2x-4)^2$$

$$\therefore 3(2x-4) > -2(2x-4)^2$$

$$\therefore 2(2x-4)^2 + 3(2x-4) > 0$$

$$\therefore (2x-4)[2(2x-4) + 3] > 0$$

$$\therefore (2x-4)(4x-5) > 0$$



$$\therefore x < 5/4 \text{ or } x > 2$$

$$(d) - \left[\sin^{-1} \left(\frac{x}{\sqrt{2}} \right) \right]_{-1}^1$$

$$= - \left[\sin^{-1} \left(\frac{1}{\sqrt{2}} \right) - \sin^{-1} \left(\frac{-1}{\sqrt{2}} \right) \right]$$

$$= - \left[\frac{\pi}{4} + \frac{\pi}{4} \right]$$

$$= - \frac{\pi}{2}$$

$$(e) \cos 2x = 1 - 2 \sin^2 x$$

$$\therefore \cos 8x = 1 - 2 \sin^2 4x$$

$$\therefore \int_0^{\pi/12} 2 \sin^2 4x \, dx = \int_0^{\pi/12} 2 \times \frac{1}{2} (1 - \cos 8x) \, dx$$

$$= \int_0^{\pi/12} 1 - \cos 8x \, dx$$

$$= \left[x - \frac{1}{8} \sin 8x \right]_0^{\pi/12}$$

$$= \left(\frac{\pi}{12} - \frac{1}{8} \sin \left(\frac{2\pi}{3} \right) \right) - \left(0 - \frac{1}{8} \sin 0 \right)$$

$$= \frac{\pi}{12} - \frac{1}{8} \times \frac{\sqrt{3}}{2}$$

$$= \frac{\pi}{12} - \frac{\sqrt{3}}{16}$$

Question 2

$$(a) u = 3x^4 - 3$$

$$\therefore \frac{du}{dx} = 12x^3 \quad \therefore \frac{du}{2} = 6x^3 \, dx$$

$$\therefore \int (6x^3 \sqrt{3x^4 - 3})^3 \, dx = \int \frac{1}{2} \sqrt{u^3} \, du$$

$$= \frac{1}{2} \int u^{3/2} \, du$$

$$= \frac{1}{2} \times \frac{2}{5} u^{5/2} + C$$

$$= \frac{1}{5} (3x^4 - 3)^{5/2} + C \quad \text{or} \quad \frac{\sqrt{3x^4 - 3}^5}{5} + C$$

$$(b) (i) \text{ Domain: } -1 \leq \frac{x}{3} \leq 1$$

$$\therefore -3 \leq x \leq 3$$

$$\text{Range: } -\pi \leq y \leq \pi$$

$$\therefore A(3, \pi)$$

$$(ii) \frac{dy}{dx} = 2x \sin^{-1} \left(\frac{x}{3} \right) + 2x \times \frac{1}{\sqrt{1 - \frac{x^2}{9}}} + 2 \times \frac{1}{2} (9-x^2)^{-1/2}$$

$$= 2 \sin^{-1} \left(\frac{x}{3} \right) + \frac{2x}{\sqrt{9-x^2}} - \frac{2x}{(9-x^2)^{1/2}}$$

$$= 2 \sin^{-1} \left(\frac{x}{3} \right) + \frac{2x}{\sqrt{9-x^2}} - \frac{2x}{\sqrt{9-x^2}}$$

$$= 2 \sin^{-1} \left(\frac{x}{3} \right)$$

Q2 (cont.)

$$\begin{aligned}
 \text{(b)(iii) Area} &= \int_0^3 2 \sin^{-1}\left(\frac{x}{3}\right) dx \\
 &= \left[2x \sin^{-1}\left(\frac{x}{3}\right) + 2\sqrt{9-x^2} \right]_0^3 \\
 &= \left[2 \times 3 \sin^{-1}(1) + 2\sqrt{9-9} \right] \\
 &\quad - \left[2 \times 0 \sin^{-1}0 + 2\sqrt{9-0} \right] \\
 &= \left(6 \times \frac{\pi}{2} + 0 \right) - (0 + 2 \times 3) \\
 &= (3\pi - 6) \text{ units}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)(i)} \quad \sqrt{3} \sin t + \cos t &= R \sin(t + \alpha) \\
 &= R \sin t \cos \alpha + R \cos t \sin \alpha \\
 \therefore \sqrt{3} &= R \cos \alpha \quad \therefore \cos \alpha = \frac{\sqrt{3}}{R} \\
 1 &= R \sin \alpha \quad \therefore \sin \alpha = \frac{1}{R}
 \end{aligned}$$

$$\begin{aligned}
 R &= \sqrt{3+1} \\
 R &= 2 \\
 \tan \alpha &= \frac{1}{\sqrt{3}} \\
 \therefore \alpha &= \frac{\pi}{6}
 \end{aligned}$$

$$\therefore \sqrt{3} \sin t + \cos t = 2 \sin\left(t + \frac{\pi}{6}\right)$$

$$\text{(ii)} \quad \sqrt{3} \sin t + \cos t = \sqrt{3}$$

$$\therefore 2 \sin\left(t + \frac{\pi}{6}\right) = \sqrt{3}$$

$$\therefore \sin\left(t + \frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

$$\therefore t + \frac{\pi}{6} = \frac{\pi}{3}, \pi - \frac{\pi}{3}, 2\pi + \frac{\pi}{3}, 3\pi - \frac{\pi}{3}, \dots$$

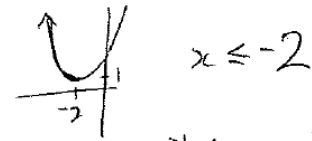
$$\therefore t = \frac{\pi}{3} - \frac{\pi}{6}, \frac{2\pi}{3} - \frac{\pi}{6}, \frac{7\pi}{3} - \frac{\pi}{6}, \dots$$

$$t = \frac{\pi}{6}, \frac{3\pi}{6}, \frac{13\pi}{6}$$

$$\therefore t = \frac{\pi}{6}, \frac{\pi}{2}$$

Question 3

$$\text{(a)(i)} \quad (x+2)^2 = (y-1) \quad \text{Vertex}(-2, 1)$$



$$\text{(ii) Domain of } f^{-1}(x): x \geq 1$$

$$\begin{aligned}
 \text{(b) General term} &= {}^{15}C_r (x^4)^{15-r} \left(\frac{3}{x^2}\right)^r \\
 &= {}^{15}C_r x^{60-4r} 3^r (x^{-2})^r \\
 &= {}^{15}C_r 3^r x^{60-6r}
 \end{aligned}$$

$$\text{constant term when } 60-6r=0$$

$$\therefore r=10$$

$$\therefore \text{Constant term is } {}^{15}C_{10} 3^{10}$$

$$\text{(c)(i)} \quad T-15 = Ae^{-kt}$$

$$\frac{dT}{dt} = 0 + -kAe^{-kt}$$

$$= -k(T-15)$$

$$\text{(ii) When } t=0, T=88$$

$$\therefore 88 = 15 + Ae^0$$

$$\therefore A = 88 - 15 = 73$$

$$\text{when } t=11, T=55$$

$$\therefore 55 = 15 + 73e^{-11k}$$

$$\therefore 40 = 73e^{-11k}$$

$$\therefore \frac{40}{73} = e^{-11k}$$

$$\therefore -11k = \log_e\left(\frac{40}{73}\right)$$

$$\therefore k = -\frac{1}{11} \ln\left(\frac{40}{73}\right)$$

$$\text{(iii)} \quad 33 = 15 + 73e^{-kt}$$

$$\therefore 18 = 73e^{-kt}$$

$$\therefore \frac{18}{73} = e^{-kt}$$

$$\therefore -kt = \ln\left(\frac{18}{73}\right)$$

$$\therefore t = \frac{\ln\left(\frac{18}{73}\right)}{-\frac{1}{11} \ln\left(\frac{40}{73}\right)} = 25.6 \text{ min} \approx 25 \text{ min } 36 \text{ sec}$$

Q3 (cont.)

(d) Prove true for $n=1$

$$\text{LHS} = (2-1)^2 = 1 \quad \text{RHS} = \frac{1}{3} \times 1 \times (2-1) \times (2+1) = 1$$

$\therefore$ True for $n=1$

Assume true for $n=k$

$$\text{i.e. } 1^2 + 3^2 + \dots + (2k-1)^2 = \frac{1}{3} k(2k-1)(2k+1)$$

Now prove true for $n=k+1$

$$\text{RHS} = \frac{1}{3} k(2k-1)(2k+1) + (2(k+1)-1)^2$$

$$= \frac{1}{3} k(2k-1)(2k+1) + (2k+1)^2$$

$$= \frac{1}{3} (2k+1) [k(2k-1) + 3(2k+1)]$$

$$= \frac{1}{3} (2k+1) (2k^2 + 5k + 3)$$

$$= \frac{1}{3} (2k+1) (2k+3) (k+1)$$

$$= \frac{1}{3} (k+1) (2(k+1)-1) (2(k+1)+1)$$

$\therefore$ Statement is true for $n=k+1$

Since statement is true for $n=1$

and $n=k+1$, then it is true for $n=2, 3, 4, \dots$ i.e. for all integers $n \geq 1$

Question 4

(a) (i) Gradient of SW = p , $y\text{-int} = a$

$\therefore$ Eqn of SW is $y = px + a$

$$\left. \begin{aligned} x + py &= ap^3 + 2ap \\ y &= px + a \end{aligned} \right\}$$

$$\therefore x + p(px + a) = ap^3 + 2ap$$

$$\therefore x + p^2x + ap = ap^3 + 2ap$$

$$\therefore p^2x + x = ap^3 + ap$$

$$\therefore x(p^2 + 1) = ap(p^2 + 1)$$

$$\therefore x = ap$$

$$y = p \times ap + a$$

$$= ap^2 + a \quad \therefore W(ap, ap^2 + a)$$

$$(ii) ap = x \quad \therefore p = \frac{x}{a}$$

$$y = ap^2 + a$$

$$\therefore y = a \left(\frac{x}{a} \right)^2 + a$$

$$y = \frac{x^2}{a} + a \quad (\text{OR } x^2 = a(y-a))$$

$$(b) (i) x = 2 \cos \left(3t - \frac{\pi}{6} \right)$$

$$\dot{x} = -2 \times 3 \sin \left(3t - \frac{\pi}{6} \right)$$

$$\ddot{x} = -2 \times 9 \cos \left(3t - \frac{\pi}{6} \right)$$

$$= -9 \times 2 \cos \left(3t - \frac{\pi}{6} \right)$$

$$\ddot{x} = -9x \text{ which is in the form}$$

$$\ddot{x} = -n^2x \text{ where } n=3$$

$\therefore$ SHM

$$(ii) \text{Period} = \frac{2\pi}{3}$$

$$(iii) \sqrt{3} = 2 \cos \left(3t - \frac{\pi}{6} \right)$$

$$\cos \left(3t - \frac{\pi}{6} \right) = \frac{\sqrt{3}}{2}$$

$$\therefore 3t - \frac{\pi}{6} = \frac{\pi}{6}, \dots$$

$$\therefore 3t = \frac{\pi}{3}, \dots$$

$$\therefore t = \frac{\pi}{9}, \dots$$

$$\text{when } t = \frac{\pi}{9}, v = -6 \sin \left(3 \times \frac{\pi}{9} - \frac{\pi}{6} \right) = -6 \sin \left(\frac{\pi}{3} - \frac{\pi}{6} \right)$$

$$= -6 \sin \left(\frac{\pi}{6} \right)$$

$$= -6 \times \frac{1}{2}$$

$$= -3 \text{ m/s}$$

$$(c) (i) \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -2e^{-x}$$

$$\therefore \frac{1}{2} v^2 = 2e^{-x} + c$$

$$\text{when } x=0, v=2$$

$$\therefore \frac{4}{2} = 2e^0 + c$$

$$\therefore 2 = 2 + c \quad \therefore c = 0$$

$$\therefore \frac{1}{2} v^2 = 2e^{-x}$$

$$\therefore v^2 = 4e^{-x}$$

$$\therefore v = \pm \sqrt{4e^{-x}}$$

$$\text{But when } x=0, v=2$$

$$\therefore v = +2\sqrt{e^{-x}}$$

$$\therefore v = 2(e^{-x})^{1/2} \quad \therefore v = 2e^{-x/2}$$

Q4 (cont.)

$$(i) \frac{dx}{dt} = 2e^{\frac{x}{2}}$$

$$\therefore \frac{dt}{dx} = \frac{1}{2e^{\frac{x}{2}}} = \frac{1}{2}e^{-\frac{x}{2}}$$

$$\therefore t = \frac{1}{2} \times 2e^{\frac{x}{2}} + C$$

$$t = e^{\frac{x}{2}} + C$$

When $t=0, x=0$

$$0 = e^0 + C \therefore C = -1$$

$$\therefore t = e^{\frac{x}{2}} - 1$$

$$\therefore t+1 = e^{\frac{x}{2}}$$

$$\therefore \frac{x}{2} = \log_e(t+1)$$

$$\therefore x = 2 \ln(t+1)$$

Question 5

$$(a) (i) x = \frac{vt}{\sqrt{2}} \text{ and } y = \frac{vt}{\sqrt{2}} - 5t^2$$

$$x = 3 \text{ gives } 3 = \frac{vt}{\sqrt{2}}$$

$$\text{And } y = 0.5 \therefore t = \frac{3\sqrt{2}}{v}$$

$$\therefore 0.5 = \frac{v}{\sqrt{2}} \times \frac{3\sqrt{2}}{v} - 5 \times \left(\frac{3\sqrt{2}}{v}\right)^2$$

$$0.5 = 3 - 5 \times \frac{9 \times 2}{v^2}$$

$$\therefore -2.5 = -\frac{90}{v^2}$$

$$\therefore 2.5v^2 = 90$$

$$\therefore v^2 = 36$$

$$\therefore v = 6 \text{ m/s (since } v > 0)$$

(ii) Max height when $\dot{y} = 0$

$$y = \frac{6t}{\sqrt{2}} - 5t^2$$

$$\therefore \dot{y} = \frac{6}{\sqrt{2}} - 10t$$

$$0 = \frac{6}{\sqrt{2}} - 10t$$

$$10t = \frac{6}{\sqrt{2}}$$

$$t = \frac{3}{5\sqrt{2}}$$

(a)(ii) (cont.) When $t = \frac{3}{5\sqrt{2}}$

$$y = \frac{6}{\sqrt{2}} \times \frac{3}{5\sqrt{2}} - 5 \times \left(\frac{3}{5\sqrt{2}}\right)^2$$

$$= \frac{18}{5 \times 2} - 5 \times \frac{9}{25 \times 2}$$

$$= 1.8 - 0.9$$

$$= 0.9 \text{ m}$$

i.e. 0.4 m (or 40 cm) above basket

(iii) $x = 3$ m basket

$$\therefore 3 = \frac{6t}{\sqrt{2}} \therefore t = \frac{3\sqrt{2}}{6} = \frac{\sqrt{2}}{2} \text{ sec}$$

$$\text{So } \dot{x} = \frac{6}{\sqrt{2}} \text{ and } \dot{y} = \frac{6}{\sqrt{2}} - \frac{10 \times \sqrt{2}}{2}$$

$$= \frac{6\sqrt{2}}{2} = 3\sqrt{2} \quad = \frac{6\sqrt{2}}{2} - 5\sqrt{2}$$

$$= -2\sqrt{2}$$

$$\therefore v^2 = (2\sqrt{2})^2 + (3\sqrt{2})^2$$

$$v^2 = 4 \times 2 + 9 \times 2$$

$$v^2 = 26$$

$$\therefore \text{Speed} = \sqrt{26} \text{ m/s}$$

$$(b) (i) f\left(\frac{a+b}{2}\right) = \left(\frac{a+b}{2} - a\right)\left(\frac{a+b}{2} - b\right)\left(\frac{a+b}{2} - c\right)$$

$$= \left(\frac{a+b-2a}{2}\right)\left(\frac{a+b-2b}{2}\right)\left(\frac{a+b-2c}{2}\right)$$

$$= \frac{1}{2}(b-a)\frac{1}{2}(a-b)\frac{1}{2}(a+b-2c)$$

$$= \frac{1}{8}(b-a)(a-b)(a+b-2c)$$

$$= -\frac{1}{8}(a-b)(a-b)(a+b-2c)$$

$$= -\frac{1}{8}(a-b)^2(a+b-2c)$$

$$(ii) f'\left(\frac{a+b}{2}\right) = \left(\frac{a+b}{2} - a\right)\left(\frac{a+b}{2} - b\right) + \left(\frac{a+b}{2} - a\right)\left(\frac{a+b}{2} - c\right)$$

$$+ \left(\frac{a+b}{2} - b\right)\left(\frac{a+b}{2} - c\right)$$

$$\therefore f'\left(\frac{a+b}{2}\right) = \left(\frac{a+b-2a}{2}\right)\left(\frac{a+b-2b}{2}\right) + \left(\frac{a+b-2a}{2}\right)\left(\frac{a+b-2c}{2}\right)$$

$$+ \left(\frac{a+b-2b}{2}\right)\left(\frac{a+b-2c}{2}\right)$$

$$= \frac{1}{4}(b-a)(a-b) + \frac{1}{4}(b-a)(a+b-2c) + \frac{1}{4}(a-b)(a+b-2c)$$

$$= -\frac{1}{4}(a-b)(a-b) - \frac{1}{4}(a-b)(a+b-2c) + \frac{1}{4}(a-b)(a+b-2c)$$

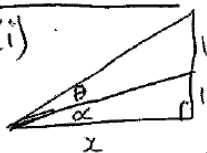
$$= -\frac{1}{4}(a-b)^2$$

(Q5) (cont.)

$$\begin{aligned} \text{(b)(iii)} \quad m &= \frac{(-\frac{1}{8}(a-b)^2(a+b-2c)-0) \times \frac{2}{2}}{(\frac{a+b}{2}-c)} \\ &= \frac{-\frac{1}{4}(a-b)^2(a+b-2c)}{(a+b-2c)} \\ &= -\frac{1}{4}(a-b)^2 \end{aligned}$$

(iv) Since the gradients in parts (ii) & (iii) are equal, the tangent at M must pass through $x=c$

Question 6

(a)(i)  $\tan \alpha = \frac{1}{x}$
 $\therefore \alpha = \tan^{-1}(\frac{1}{x})$

$$\begin{aligned} \tan(\theta + \alpha) &= \frac{2}{x} \\ \therefore \theta + \alpha &= \tan^{-1}(\frac{2}{x}) \\ \therefore \theta &= \tan^{-1}(\frac{2}{x}) - \alpha \\ \therefore \theta &= \tan^{-1}(\frac{2}{x}) - \tan^{-1}(\frac{1}{x}) \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \theta &= \tan^{-1}(2x^{-1}) - \tan^{-1}(x^{-1}) \\ \therefore \frac{d\theta}{dx} &= \frac{-2x^{-2}}{1 + \frac{4}{x^2}} - \frac{-x^{-2}}{1 + \frac{1}{x^2}} \\ &= \frac{-\frac{2}{x^2}}{1 + \frac{4}{x^2}} + \frac{\frac{1}{x^2}}{1 + \frac{1}{x^2}} \\ &= \frac{-2}{x^2+4} + \frac{1}{x^2+1} \end{aligned}$$

Maximum when $\frac{d\theta}{dx} = 0$

$$\therefore 0 = \frac{-2}{x^2+4} + \frac{1}{x^2+1}$$

$$\therefore \frac{2}{x^2+4} = \frac{1}{x^2+1}$$

$$\therefore 2x^2+2 = x^2+4$$

$$\therefore x^2 = 2$$

$$\therefore x = \sqrt{2} \text{ or } -\sqrt{2}$$

Test $x = \sqrt{2}$ (since $x > 0$)

x	1	$\sqrt{2}$	2
$\frac{d\theta}{dx}$	$\frac{1}{10}$	0	$-\frac{1}{20}$

$\therefore$ Max when $x = \sqrt{2}$

$$\begin{aligned} \text{(iii) Max viewing angle} &= \tan^{-1}(\frac{2}{\sqrt{2}}) - \tan^{-1}(\frac{1}{\sqrt{2}}) \\ &= \tan^{-1}(\sqrt{2}) - \tan^{-1}(\frac{1}{\sqrt{2}}) \\ &= 54^\circ 44' 8.2'' - 35^\circ 15' 51.8'' \\ &= 19^\circ 28' 16.39'' \\ &\approx 19^\circ \text{ (nearest degree)} \end{aligned}$$

$$\text{(b)(i)} \quad \tan \alpha = \frac{h}{WB} \quad \therefore WB = \frac{h}{\tan \alpha} = h \cot \alpha$$

$$\text{Also, } \tan \alpha = \frac{h}{BX} \quad \therefore BX = \frac{h}{\tan \alpha} = h \cot \alpha$$

$\therefore \triangle BWX$ is isosceles ($WB = BX$, i.e. 2 equal sides)

$$\begin{aligned} \text{(ii)} \quad BM &= \sqrt{h^2 \cot^2 \alpha - 20^2} \\ &= \sqrt{h^2 \cot^2 \alpha - 400} \\ \text{OR } BM &= \sqrt{h^2 \cot^2 \beta - 3600} \end{aligned}$$

$$\text{(iii)} \quad \tan \beta = \frac{h}{BY} \quad \therefore BY = h \cot \beta$$

$$\begin{aligned} BY^2 &= BM^2 + MY^2 \\ MY &= 20 + 40 = 60 \end{aligned}$$

$$h^2 \cot^2 \beta = (h^2 \cot^2 \alpha - 400) + 60^2$$

$$h^2 \cot^2 \beta = h^2 \cot^2 \alpha - 400 + 3600$$

$$h^2 \cot^2 \beta - h^2 \cot^2 \alpha = 3200$$

$$h^2 (\cot^2 \beta - \cot^2 \alpha) = 3200$$

$$h^2 = \frac{3200}{(\cot^2 \beta - \cot^2 \alpha)}$$

$$\therefore h = \frac{\sqrt{3200}}{\sqrt{\cot^2 \beta - \cot^2 \alpha}} \quad (\text{since } h > 0)$$

$$\therefore h = \frac{40\sqrt{2}}{\sqrt{\cot^2 \beta - \cot^2 \alpha}}$$

Question 7

(a)(i) $7! = {}^7P_7 = 5040$ ways

(ii) $\left. \begin{array}{l} GG\ BBBBB \\ BG\ BBBBB \\ BB\ GBBBB \\ BBB\ GBB \\ BBBB\ GB \\ BBBBBA \end{array} \right\}$ Each of these can be arranged $5! \times 2!$ ways
 $\therefore \text{Total} = 6! \times 2! = 1440$ ways

(iii) $\left. \begin{array}{l} GBBB\ B \\ BG\ BB\ B \\ BB\ G\ BB \\ BBB\ G\ B \\ BBBB\ G \end{array} \right\}$ Each of these can be arranged $5! \times 2!$ ways
 $\therefore \text{Total} = 4 \times 5! \times 2! = 960$ ways

(b)(i) $\log_e(x^2)$ has a domain of all real x but $x \neq 0$

$2 \log_e x$ only has a domain of $x > 0$, so half of the curve would be deleted/lost

(ii) $x \neq 0$ since $\log_e 0$ is undefined
 And since $\log_e 1 = 0$ and $\frac{x}{0}$ is undefined
 thus $x \neq \pm 1$

$\therefore$ Domain is all real x but $x \neq 0, x \neq \pm 1$

(iii) $\frac{dy}{dx} = \frac{(\ln x^2) \times 1 - x \times \frac{2x}{x^2}}{(\ln x^2)^2}$
 $= \frac{\ln x^2 - 2}{(\ln x^2)^2}$

(iv) stat pts when $\frac{dy}{dx} = 0$

$\therefore 0 = \frac{\ln x^2 - 2}{(\ln x^2)^2}$

$\therefore 2 = \ln x^2$

$\therefore x^2 = e^2$

$\therefore x = \pm \sqrt{e^2}$

$\therefore x = e$ or $x = -e$

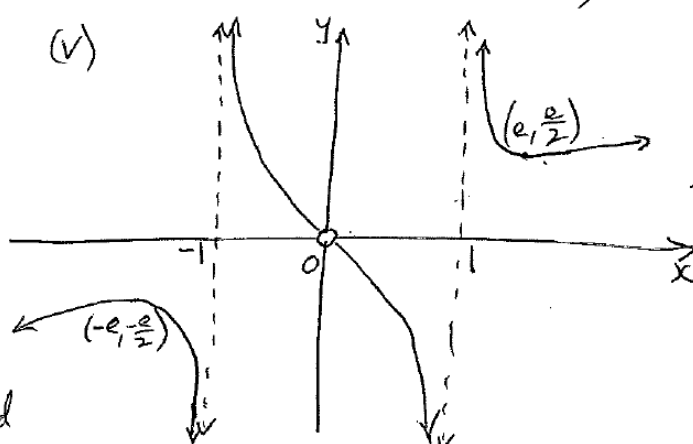
$y = \frac{e}{2} \quad y = -\frac{e}{2}$

x	-3	-e	-2
y'	$\frac{\ln 9 - 2}{(\ln 9)^2} \approx 0.04$	0	$\frac{\ln 4 - 2}{(\ln 4)^2} \approx -0.32$

$\therefore$ Max turning pt @ $(-e, -\frac{e}{2})$

x	2	e	3
y'	$\frac{\ln 4 - 2}{(\ln 4)^2} \approx -0.32$	0	$\frac{\ln 9 - 2}{(\ln 9)^2} \approx 0.04$

$\therefore$ Min turning pt @ $(e, \frac{e}{2})$



when $x = \frac{1}{2}, y = \frac{0.5}{\ln 0.25} < 0$

when $x = -\frac{1}{2}, y = \frac{-0.5}{\ln 0.25} > 0$